«LangGraph子图组装者»
MailQueryWorkflow
-MailAgent _agent
-MailQueryService _query_service
-ConversationArtifactService _artifacts
+build() : CompiledStateGraph
-_plan(state) : dict
-_query(state) : dict
读取交接要求
plan_query
写入State.plan
query
构造可信结果摘要
构造有序候选卡片
persist
«已校验交接契约»
AgentHandoff
+domain mail
+intent query_mail
+instruction str
+target ResourceReference
«领域规划器»
MailAgent
-OpenAICompatibleClient _llm_client
+plan_query(user_content, handoff) : MailQueryPlan
«查询业务与审计编排»
MailQueryService
-Database _database
-ToolGateway _tools
-MailQueryRepository _repository
-AgentRunRepository _runs
-ToolCallRepository _tool_calls
-RunEventPublisher _events
+query(run, plan) : tuple
«安全结果摘要»
ToolResultSummary
+status str
+tool_name str
+summary dict
«前端卡片数据»
ConversationArtifact
+artifact_type mail_candidates
+payload dict
ConversationArtifactService
+persist(run, artifacts) : list
LLM生成并校验
固定mail.messages.list
prepare及invoke
校验并去除额外字段
同步快照取得email_id
matches确定性筛选
调用记录及连接
步骤和进度事件
progress_updated及
tool_preview
保存Artifact
写artifacts_ready
事务提交后发布
ToolInvocation
+tool_name str
+arguments dict
«Pydantic计划契约»
MailQueryPlan
+version int
+action list_filter_clarify
+keywords list
+sender_contains str
+start_at datetime
+end_at datetime
+limit int
+clarification_question str
+matches(message) : bool
«工具安全入口»
ToolGateway
-ToolRegistry _registry
-ProviderConnectionRepository _connections
+get_definition(tool_name) : ToolDefinition
+prepare(session, invocation) : PreparedToolInvocation
+invoke(prepared) : ToolResult
«白名单元数据»
MailListMessage
+provider_message_id str
+subject str
+sender_email str
+received_at datetime
+source_version str
«快照持久化»
MailQueryRepository
+upsert(session, run, connection_id, messages) : list
«工具调用审计»
ToolCallRepository
+create_running() : UUID
+bind_connection()
+succeed()
+fail()
«任务步骤与事件»
AgentRunRepository
+create_running_tool_step() : UUID
+finish_step()
+append_event() : RunEvent
ConversationArtifactRepository
+upsert_for_run(session, run, artifacts) : list
«Redis事件发布器»
RunEventPublisher
+publish_after_commit(event)
查询注册关系
校验租户用户及mail.read
email_messages
tool_calls
run_steps及run_events
conversation_message_artifacts
«已注册工具表»
ToolRegistry
+register(definition, handler)
+get(tool_name) : RegisteredTool
ProviderConnectionRepository
+find_active_connection() : ProviderConnection
«database»
PostgreSQL
email_messages
tool_calls
run_steps
run_events
conversation_message_artifacts
get返回
invoke执行
RegisteredTool
+definition ToolDefinition
+handler ToolHandler
读取response_payload
handler引用
«邮件工具适配器»
EmailMCPToolHandler
-EmailMCPClient _client
+invoke(context, arguments) : ToolResult
call
«MCP客户端»
EmailMCPClient
+call(tool_name, context, arguments) : ToolResult
MCP请求
返回结构化响应
«外部工具服务»
EmailMCPServer
+mail_messages_list()
+mail_messages_get()
+mail_messages_send()
«工具响应»
ToolResult
+response_payload dict
+output_summary dict